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# Computational Investigation of the Antioxidant Activity of Novel Ascorbic Acid Derivatives: Thermochemical, kinetic, Drug-likeness and ADMET Studies

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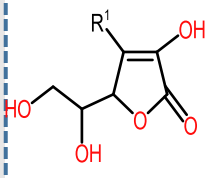
## Introduction




**R1= $NH_2$, ONa, AG**

-Study the antioxidant activity of vitamin C using thermochemical descriptors.
- Design new derivatives of VC (AA-$NH_2$, AA-AG, AA-O-Na) by electron donor substitution of OH and study of their thermodynamic and kinetic aspects.
- Predicting the druglikeness and ADMET properties of vitamin C and its derivatives.

## Computational methodology

- All computations were carried out using Gaussian 09 software at the SMD//M05-2X /6-31+G(d,p) level of theory.
- The main antioxidant mechanisms considered in this work are **HAT, SPLET, SET-PT**
- The rate constants (*k*) were calculated according to the transition state theory:

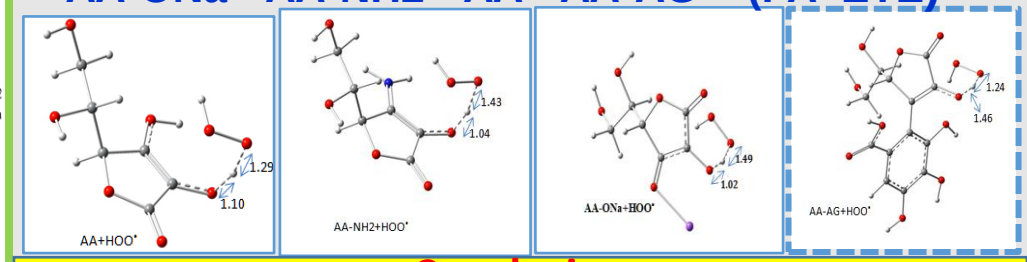
$$k = \sigma\chi(T) + \frac{k_B T}{h} e^{-\Delta G^{\neq}/RT}$$

## Results

| | HAT | | | SPLET | SET-PT | | |
|---|---|---|---|---|---|---|---|
| | BDE (kJ /mol) | PA (kJ /mol) | ETE (kJ /mol) | PA+ETE (kJ/mol) | IP (kJ/mol) | PDE (kJ/mol) | IP+PDE (kJ/mol) |
| AA | 331.12 | 142.06 | 390.27 | **532.33** | 532.17 | 0.15 | 532.33 |
| AA-$NH_2$ | 298.41 | 155.61 | 344.00 | **499.61** | 471.71 | 27.90 | 499.61 |
| AA-ONa | 288.47 | 175.84 | 313.82 | **489.66** | 440.88 | 48.79 | 489.66 |
| AA-AG | 339.29 | 129.23 | 411.26 | **540.49** | 532.57 | 7.93 | 540.49 |

➢ The thermodynamically favored HAT mechanism in the gas phase, on the other hand, the SPLET mechanism is the most favored in water.
➢ The designed derivatives have better antioxidant activity than vitamin C.

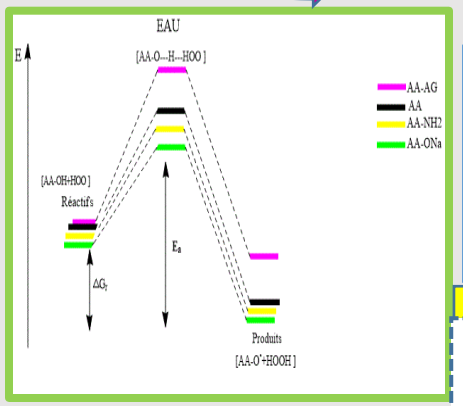
**AA-ONa >AA-NH2 >AA> AA-AG (PA+ETE)**




## Kinetic results

➢ Compounds AA-Ona and AA-$NH_2$ are kinetically favored to react with HOO·

➢ The results obtained in the kinetic part are in good agreement with the results obtained in the thermodynamic part.




## Drug-likeness , ADMET

| | AA | AA-$NH_2$ | AA-ONa | AA-AG |
|---|---|---|---|---|
| **Lipinski** | Yes | Yes | Yes | Yes |
| **Veber** | Yes | Yes | Yes | No |
| **Egan** | Yes | Yes | Yes | No |

The predicted pharmacokinetic properties show that ascorbic acid and its two designed derivatives AA-Ona and AA-$NH_2$ are non-toxic and could be easily absorbed, distributed, metabolized in the human body.

## Conclusion

- The Compound AA-ONa is the most **potent** antioxidant in both media.
- The compound AA-$NH_2$ is a good scavenger of the hydroperoxyl radical in the aqueous phase.

### References

Liu, Yuyang, Chuanqun Liu, etJianjun Li. « Comparison of Vitamin C and Its Derivative Antioxidant Activity: Evaluated by Using Density Functional Theory ». *ACS Omega 2020*